

AI in Cybersecurity and Cloud Computing

Lesson 2

AI in Cybersecurity: How It Is Used, and the Data Behind It

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Learning Objective

- Define the basic terms of cybersecurity, explain how AI is used to defend against the main attacks, and describe the security data (logs) that AI learns from.

Key Concepts

- Why cybersecurity needs AI
- Basic security terms: CIA, threat, vulnerability, attack, risk
- Events, alerts and incidents; the SOC and the SIEM
- AI against phishing, malware, ransomware, intrusions, account takeover, insiders and fraud
- Generative AI for defenders and attackers
- Logs and the data-processing steps

Recap

- AI is a field of computer science
- Machine learning learns the rules from examples: data + answers → model
- Learning with answers, without answers, and generative AI

Today: where AI is used in cybersecurity, and the data it needs

Why Cybersecurity Needs AI



Too much data

Millions of events a day



Too fast

Attacks spread in minutes



Always new

Attackers change tools daily



Too few people

Not enough analysts

AI filters and ranks; the analyst decides

What Is Cybersecurity? The CIA Triad

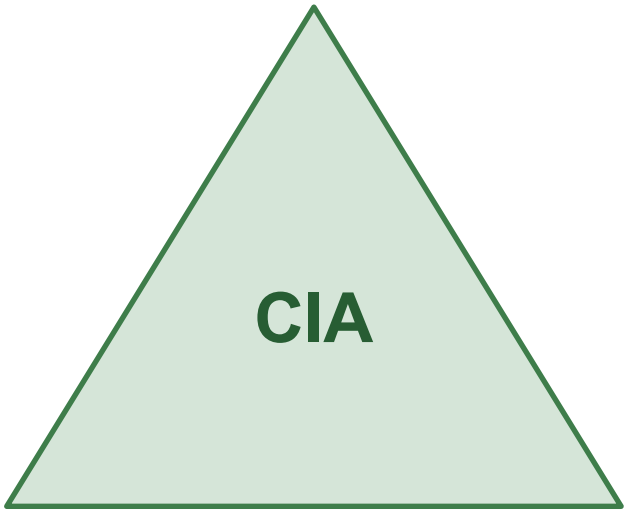
Definition **Cybersecurity:** protecting computers, networks, programs and data from attack, damage and unauthorised access.

Confidentiality

Only authorised people can read the data

Integrity

Data is not changed without permission



Availability

Systems work when they are needed

Security example: Phishing steals a password (confidentiality); ransomware encrypts files (integrity and availability); a DDoS attack takes the website down (availability).

Threat, Vulnerability, Attack and Risk

Term	Definition	Example
Threat	Anything that can cause harm to a system	A phishing gang
Vulnerability	A weakness that can be used to cause harm	An unpatched web server
Attack	An attempt to use a vulnerability to cause harm	A fake login e-mail
Risk	How likely harm is × how bad it would be	Chance and cost of losing payroll data

AI helps at every step: spotting threats, ranking vulnerabilities, detecting attacks

Events, Alerts and Incidents

Events: Millions a day: every login, connection, e-mail

Alerts: Thousands: events flagged as suspicious

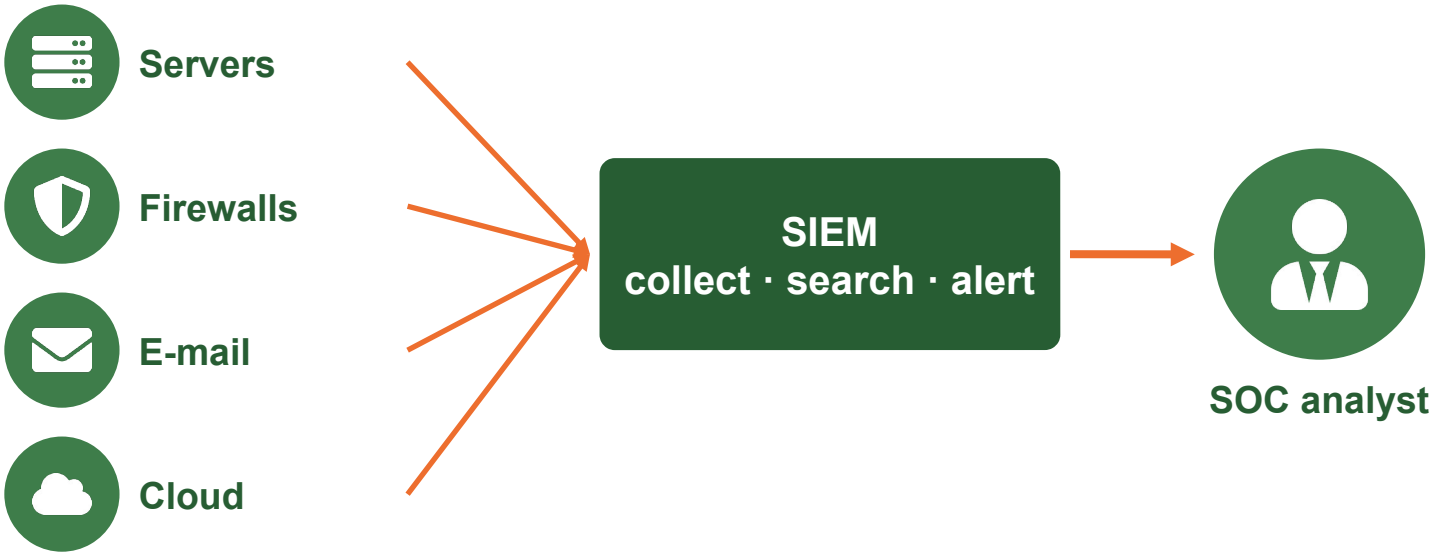
Incidents: A few: confirmed attacks

Term	Definition
Event	Anything that happens on a system and is recorded
Alert	An event flagged as possibly malicious
Incident	A confirmed security problem that needs a response

The SOC and the SIEM

Definition **SOC:** the team that monitors an organisation's systems and responds to security incidents.

Definition **SIEM:** a system that collects logs from many sources, searches them and raises alerts.



Splunk · Microsoft Sentinel · Elastic Security · Wazuh

Where AI Helps: the Defence Cycle

Govern	Identify	Protect
Policies and risk decisions	**Know your assets and weaknesses** AI: Rank which vulnerabilities to patch first	**Stop attacks before they happen** AI: Block phishing e-mails and malware

Detect	Respond	Recover
Find attacks that got in AI: Unusual logins, intrusions, insiders	**Act on incidents** AI: Summarise alerts, suggest the next step	**Return to normal** AI: Check backups are clean

Source: NIST Cybersecurity Framework 2.0 (2024)

Map: Where AI Is Used in Cybersecurity



E-mail and web

Phishing · spam · malicious domains · web attacks (WAF) · bots



Endpoints

Malware · ransomware · behaviour monitoring (EDR)



Network

Intrusions (IDS / IPS) · DDoS · IoT devices



Identity

Unusual logins · account takeover · biometrics



Data and business

Insider threats · data loss prevention · fraud



Cloud

Cloud activity monitoring · misconfigurations



Applications

Secure code review · vulnerability prioritisation



SOC operations

Alert triage · assistants · automated response



Threat intelligence

Reading reports · deepfake detection · attackers' AI

Case 1: Phishing E-mails

Definition **Phishing:** a fake message that pretends to come from someone trusted, to steal passwords or money, or to install malware.

How AI helps

- Reads many clues at once: sender, words, links, attachments
- Learns from e-mails that staff reported
- Blocks or warns before the user clicks



Learning with answers (phishing / legitimate)

Security example: “URGENT: your password expires today” from it-support@ajloun-nationa1.com: look-alike domain + urgent words + link to a fake page → blocked.

Example: Spot the Phishing E-mail

Inbox: staff@anu.edu.jo

From: IT Support <it-support@ajloun-nationa1.com> 2

Subject: URGENT: your password expires in 24 hours 1

Dear staff member,

Your university password will expire today. To avoid losing access to your e-mail and salary page, confirm your account now. 3

Confirm my account 4

link goes to: http://203.0.113.45/anu-login/verify.php

Attachment: salary_update.zip 5

1. Urgent threat: “expires in 24 hours”
2. Look-alike domain: nationa1 (digit 1)
3. Pressure: salary and e-mail access
4. Button hides an IP address, not anu.edu.jo
5. Zip attachment from “IT”

AI detector: 5 clues → phishing, blocked

Case 2: Malware

Definition **Malware:** software designed to harm a computer or its user: viruses, worms, trojans, spyware and ransomware.

How AI helps

- Classic antivirus matches fingerprints of known files
- AI looks at file features and behaviour, so it also catches new versions
- A sandbox runs the file safely and records what it does



Learning with answers (malware / clean)

Security example: A changed copy of a known trojan has a new fingerprint and passes the old antivirus, but its behaviour (keyboard capture, hidden connection) is caught by AI.

Example: A File That Pretends

Downloads



invoice.pdf

What the user sees

Real name	invoice.pdf.exe
Real type	program (.exe)
Digital signature	none
Size	312 KB

Antivirus (AI)



Threat blocked

Behaves like a trojan:
- records the keyboard
- opens a hidden connection to 198.51.100.7

Why AI caught it: Fingerprint antivirus did not know this new file. The AI model flagged it from its features (no signature, double extension) and its behaviour (keyboard capture, hidden connection).

Case 3: Ransomware

Definition **Ransomware:** malware that encrypts the victim's files and demands payment to unlock them.

How AI helps

- Learns how normal programs use files
- Flags a program that changes hundreds of files a minute
- Stops the program and isolates the PC early



Learning without answers (unusual behaviour)

Security example: A PC normally changes 3 files a minute; suddenly 400 files a minute are encrypted. The behaviour model stops the process in the first minute.

Example: Ransomware in the First Minute



10:21	Normal: 3 files changed per minute
10:27	400 files encrypted in one minute
10:27	AI flags the behaviour
10:28	Process stopped, PC isolated

Why AI helps: Because the model reacts to behaviour, it can stop new ransomware families that no signature knows yet.

Case 4: Network Intrusions (IDS / IPS)

Definition **Intrusion:** unauthorised access to a network or system. An IDS detects intrusions and alerts; an IPS can also block them.

How AI helps

- Signature-based: matches rules for known attacks (Snort, Suricata)
- Anomaly-based (AI): learns normal traffic, flags the different
- Classifies connections as normal or attack



With answers (attack types) and without answers (unusual traffic)

Security example: One address opens 250 very short connections to different ports in 2 seconds: a port scan, the first step of many attacks.

Example: The IDS Alert Console

IDS alert console: university network

Time	Severity	Source	Target	Reason	AI score
02:14	High	203.0.113.45	web-01	250 ports in 2 s (port scan)	0.97
02:16	High	203.0.113.45	srv1	120 failed SSH logins	0.95
02:31	Medium	10.0.4.18	dns-01	Many requests to new domains	0.71
03:00	Low	10.0.9.2	backup-01	Large transfer (nightly backup)	0.22

How the SOC uses it: The analyst works from the top: block 203.0.113.45 first. The 03:00 backup alert has a low score: the AI learned that this transfer happens every night.

Case 5: Account Takeover and Unusual Logins

Definition **Account takeover:** an attacker gains control of a real user's account, for example by guessing or stealing the password.

How AI helps

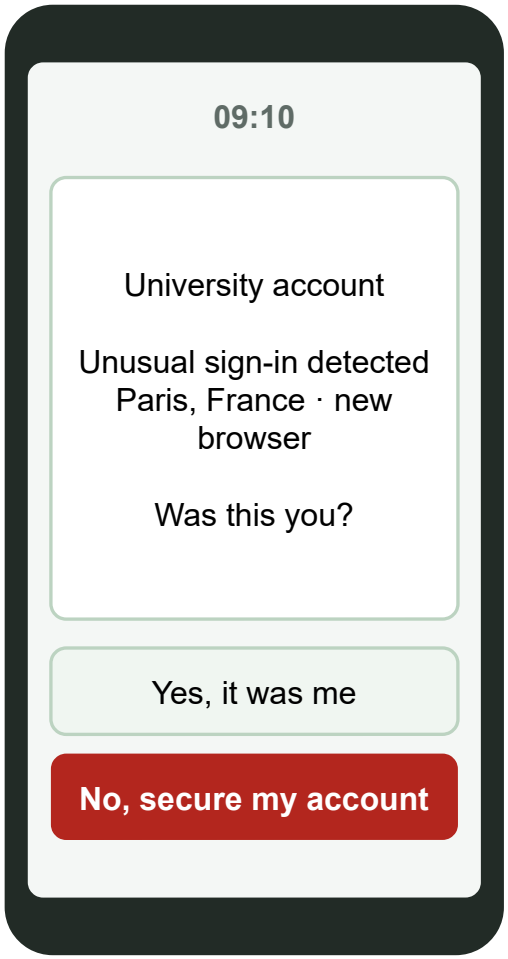
- Learns each user's normal hours, places and devices
- Flags many failed logins (brute force) and impossible travel
- Asks for a second factor or locks the account



Learning without answers (unusual logins)

Security example: Sara logs in from Amman at 09:00 and from Paris at 09:10. Nobody travels that fast: a stolen password is likely, unless it is a VPN.

Example: “Unusual Sign-In Detected”



Why the AI flagged it

1. Last sign-in: Amman at 09:00, only 10 minutes ago
2. Paris is too far for 10 minutes: impossible travel
3. New browser never used by this account
4. 6 failed passwords just before the success

Action: ask for a second factor; lock if the user says “No”

Case 6: Insider Threats

Definition **Insider threat:** a person inside the organisation who misuses their access, on purpose or by mistake.

How AI helps

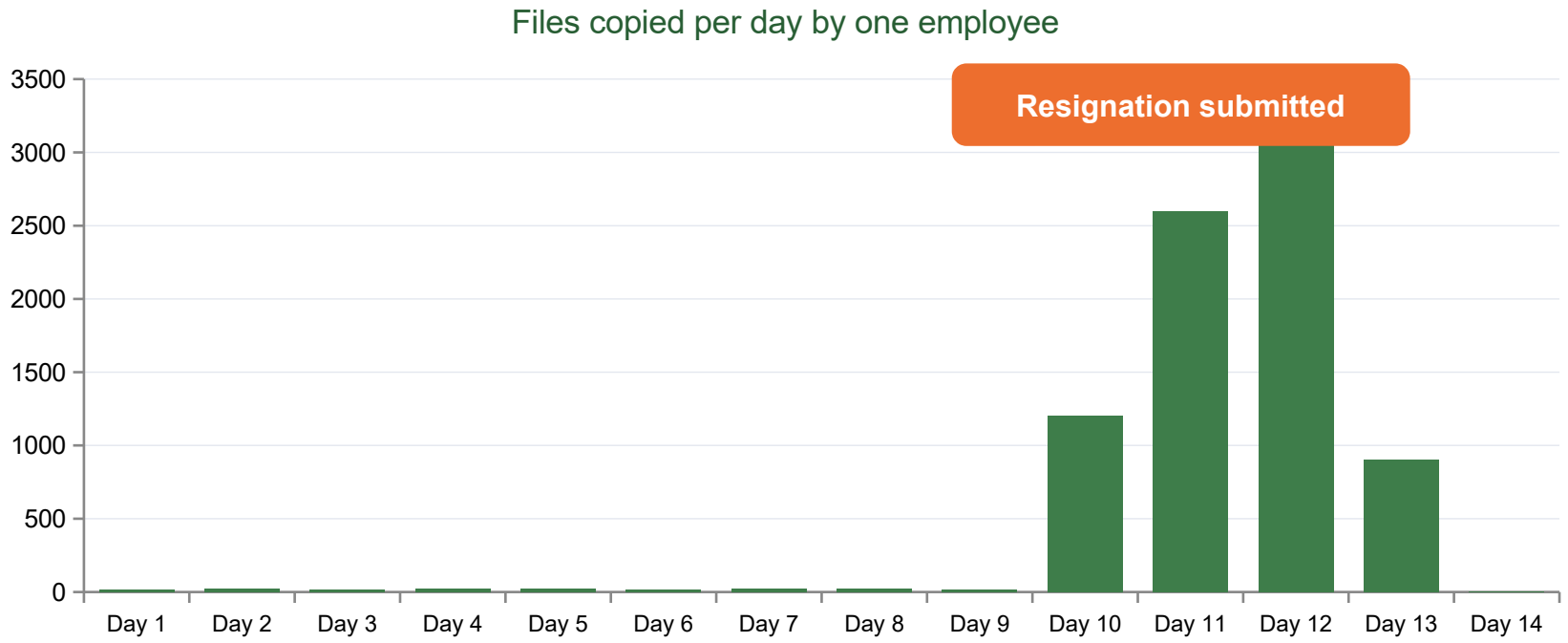
- Learns each employee's normal use of data
- Flags mass downloads, odd hours, new privileges
- Gives the analyst a ranked list, not an accusation



Learning without answers (unusual behaviour)

Security example: An employee who normally opens 20 files a day copies 5,000 files to a USB drive the week before leaving the company.

Example: An Insider's Unusual Week



How the SOC uses it: The AI does not accuse: it ranks this employee first so an analyst checks what was copied and why, following company policy.

Case 7: Fraud

Definition **Fraud:** deception for financial gain, for example using a stolen card number to buy online.

How AI helps

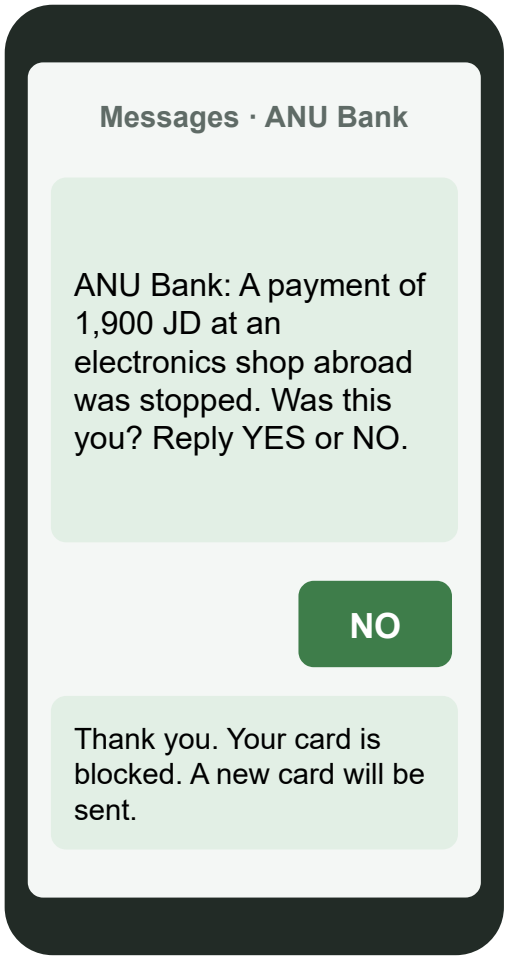
- Checks every payment in a fraction of a second
- Uses past frauds (with answers) and unusual patterns (without answers)
- Sends an SMS or blocks the payment



With answers and without answers together

Security example: A card normally used for 40 JD in Ajloun pays 1,900 JD abroad at 03:40: the bank blocks it and sends an SMS to the owner.

Example: The Bank SMS



Why the AI stopped it

Clue	Usual for this card	This payment
Amount	about 40 JD	1,900 JD
Place	Ajloun, Irbid	abroad
Time	day	03:40
Card in shop?	yes	no (online)

Score: very unusual → stop and ask the owner

Case 8: Prioritising Vulnerabilities

Definition **Vulnerability (CVE):** a known weakness in software; each public one gets an ID such as CVE-2021-44228 (Log4Shell).

How AI helps

- Thousands of new vulnerabilities are published every year
- AI predicts which ones are most likely to be exploited soon
- The team patches those first



Learning with answers (exploited / not exploited)

Security example: The free EPSS score (by FIRST) estimates how likely a vulnerability is to be exploited in the next 30 days, so the team patches the riskiest ones first.

Example: Which Vulnerability First?

Vulnerability scanner: patch priority

Priority	Vulnerability	Server	Likely to be exploited soon?	Action
1	CVE-2021-44228 (Log4Shell)	student portal	Very likely	Patch today
2	Old VPN software	remote access	Likely	Patch this week
3	Weak TLS setting	library site	Unlikely	Plan next month
4	Printer firmware	lab printer	Very unlikely	Next update

How the team uses it: The team cannot patch everything at once. The AI score plus the importance of the server tells them where to start.

More Uses (1): Web Attacks and DDoS

Web attacks and WAF

Web application attack: an attack through a website's inputs, such as SQL injection or cross-site scripting (XSS).

AI: a web application firewall (WAF) learns normal requests to the site and blocks harmful inputs and bad bots.

Example: the student-portal login box receives ' OR '1'='1 : the WAF blocks the request.

With answers (attack / normal request)

DDoS attacks

Distributed denial of service (DDoS): flooding a service with traffic from many machines so that real users cannot reach it.

AI: learns normal traffic for each hour and day, spots a sudden flood, and separates bots from real users.

Example: in registration week the portal suddenly gets 50 times its normal traffic from thousands of bots.

Without answers (unusual traffic)

More Uses (2): Malicious Domains and the Cloud



Malicious domains (DNS)

Malicious domain: a web address used for phishing or to control malware; some malware creates random-looking domains automatically.

AI: scores every domain that university computers look up and blocks the suspicious ones.

Example: a lab PC suddenly looks up kq7xz2vbt9qm.com, kq7xz2vbt9qn.com, … : machine-made names, probably malware.

With answers (bad / good domains)



Cloud security

Cloud security: protecting accounts, data and services that run in the cloud.

AI: services such as Amazon GuardDuty and Microsoft Defender for Cloud watch cloud logs for unusual activity and risky settings.

Example: a cloud key is used at night from a new country to create 20 servers: likely crypto-mining.

Without answers (unusual activity)

More Uses (3): Biometrics and Data Loss Prevention



Biometrics and liveness

Biometric authentication: proving who you are with your body: face, fingerprint or voice.

AI: recognises the face or voice, and checks liveness: is it a real person, or a photo, mask or deepfake?

Example: the phone refuses to unlock when someone holds up a printed photo of the owner.

With answers (owner / not owner)



Data loss prevention (DLP)

Data loss prevention: tools that stop sensitive data from leaving the organisation.

AI: reads documents and recognises sensitive content (student records, passport numbers, salaries).

Example: an e-mail with a file of 500 student ID numbers to a personal Gmail address is held for review.

With answers (sensitive / not)

More Uses (4): Secure Code and Threat Intelligence



Secure code review

Code vulnerability: a programming mistake that an attacker can use, such as building an SQL query from user input.

AI: AI code assistants and scanners read source code, flag risky lines and suggest a safer version.

Example: the review flags "SELECT … WHERE user='" + name + "'" and suggests a parameterised query.

With answers + generative AI



Threat intelligence

Threat intelligence: information about attackers, their methods and their indicators: bad addresses, domains and file fingerprints.

AI: reads thousands of security reports, pulls out the indicators and writes short summaries for the SOC.

Example: from a new report the AI extracts 40 bad domains and adds them to the university block list.

Generative AI + learning with answers

More Uses (5): Automated Response and IoT Devices



Automated response (SOAR)

SOAR: Security Orchestration, Automation and Response: tools that run a response plan (playbook) automatically.

AI: chooses the right playbook and runs safe steps at machine speed; risky steps wait for a human.

Example: a ransomware alert isolates the PC from the network in seconds; deleting accounts waits for an analyst.

AI + fixed playbooks



IoT devices

IoT device: a small connected device such as a camera, sensor or smart lock.

AI: learns each device's normal traffic and flags a device that suddenly behaves differently.

Example: a campus camera starts sending traffic to thousands of addresses: it joined a botnet (as in the Mirai attacks of 2016).

Without answers (unusual behaviour)

Generative AI in the SOC

Definition **Generative AI assistant:** a chat-based AI tool that writes, summarises and explains text for the analyst.

Explain an e-mail

Why does this look like phishing?

Summarise logs

500 lines → five key events

Draft a report

First draft of an incident report

Suggest a search

A query for the SIEM

Check every answer · never paste passwords or secrets

Example: Asking the SOC Assistant

SOC assistant (generative AI)

Analyst: Summarise the alerts for 203.0.113.45 from last night.

Assistant: Between 02:14 and 02:31 the address scanned 250 ports on web-01, then tried 120 SSH passwords on srv1. One login as “root” succeeded at 02:29. Recommended: block the address and reset the root password.

Assistant: The address belongs to a well-known cloud backup company.

Invented! (hallucination)

Useful summary, saves time · but check every fact before acting

Attackers Use AI Too

Definition **Deepfake:** fake audio, video or images of a real person, created with AI.



Perfect phishing

No spelling mistakes, any language



Deepfake voice

A “manager” calls and asks for a transfer

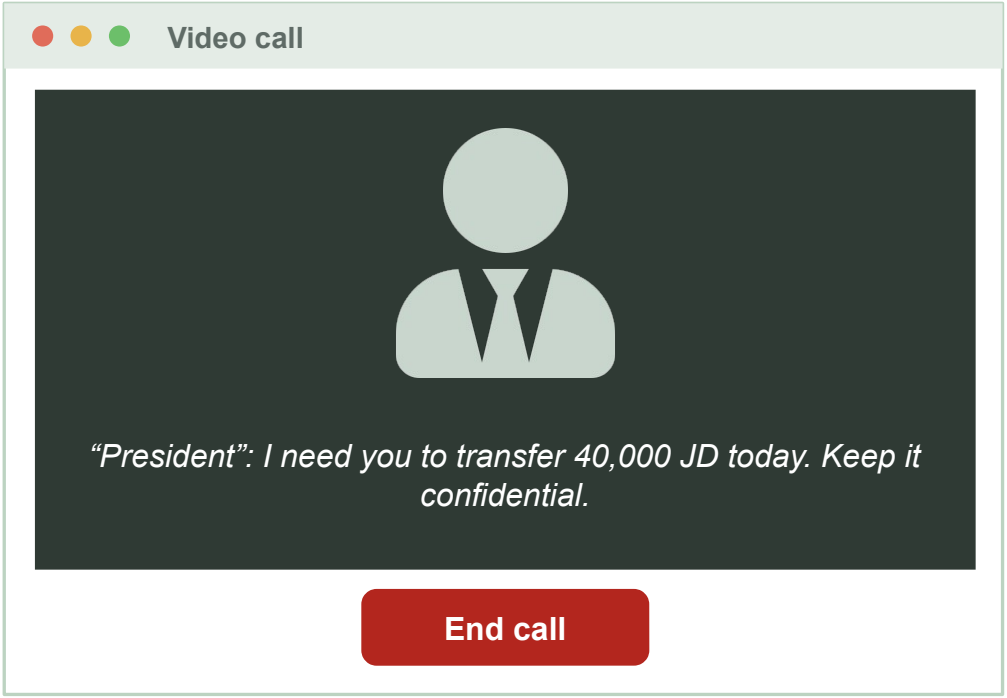


Fake websites

Copies of real login pages

Defence: verify through a second channel and train staff

Example: The Deepfake Video Call



Warning signs

1. Urgent money request
2. Asks to keep it secret
3. Skips the normal approval
4. Slight lip-sync and voice glitches

Defence: Hang up and call back on the number you already know; follow the two-person approval rule for every transfer.

The Eight Main Cases at a Glance

Case	Attack	What AI does	Kind of learning
1	Phishing	Combines clues, blocks before the click	With answers
2	Malware	Features and behaviour, not only fingerprints	With answers
3	Ransomware	Spots mass file encryption, stops it	Without answers
4	Intrusions	Classifies or flags unusual traffic	Both
5	Account takeover	Unusual logins, impossible travel	Without answers
6	Insider threat	Unusual data use	Without answers
7	Fraud	Checks every payment	Both
8	Vulnerabilities	Predicts which will be exploited	With answers

Benefits and Limits of AI in Security

Benefits	Limits
Works fast, 24 hours a day	Mistakes: false alarms and missed attacks
Handles millions of events	Attackers adapt to the detector
Finds patterns people miss	Often hard to explain (“black box”)
Lets a small team cover more	Personal data must be protected

Every Case Starts with Data

Case	Data the AI needs	Where it comes from
Phishing	E-mails with sender, links, attachments	E-mail gateway + staff reports
Malware	File features and behaviour	Antivirus, sandbox
Intrusions	Network connections (flows)	Firewall, network sensors
Logins / insiders	Who logged in, when, from where	Windows and Linux logs
Fraud	Payments: amount, place, time	Bank systems

Most security data comes from logs

Logs: The Raw Material

Definition **Log:** a record that a system writes every time something happens.

Windows event logs

4624 logon success · 4625 failure

Linux auth log

SSH logins, sudo, new users

Firewall

Allowed and blocked connections

Web server

Every page request and its status

DNS

Which domains computers look up

E-mail gateway

Sender, links, attachments

Example: Windows Logs in Event Viewer

Event Viewer › Windows Logs › Security

Level	Date and time	Event ID	Task	Account
Information	03/11/2026 02:28:01	4625	Logon failed	f.ahmad
Information	03/11/2026 02:28:09	4625	Logon failed	f.ahmad
Information	03/11/2026 02:28:15	4625	Logon failed	f.ahmad
Information	03/11/2026 02:28:22	4625	Logon failed	f.ahmad
Information	03/11/2026 02:28:30	4624	Logon success	f.ahmad

What it shows: Four failures (4625) and then a success (4624) within 30 seconds at 02:28: a guessed password is likely. This is the raw data behind Case 5.

Reading a Log Line

```
Nov 03 02:14:55 srv1 sshd[4122]: Failed password for root from 203.0.113.45
port 51522 ssh2
```

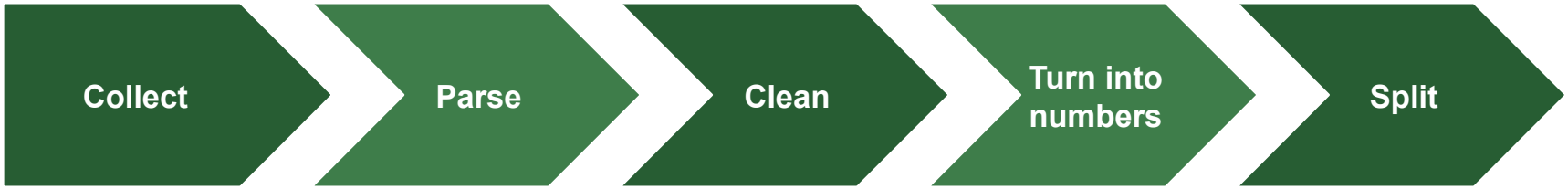
When	Which server	Which program
Nov 03 02:14:55	srv1	sshd

What happened	Which user	From where
Failed password	root	203.0.113.45

Many such lines from one address in minutes = brute-force attack

From Logs to Data the AI Can Use

Definition **Data processing:** the steps that turn raw security data into a clean table of numbers a model can learn from.



All logs in the SIEM | Line → named fields | Fix gaps and errors | Text → numbers | Learn part / test part

```
… sshd: Failed password for root from 203.0.113.45 …
```

time	event	user	source_ip
02:14:55	failed	root	203.0.113.45

One line becomes one row; each field becomes a column

Common Problems in Security Data

Problem	Example in logs	Simple fix
Missing value	Country field empty	Fill it or mark it as “unknown”
Duplicate	Same event stored twice	Keep one copy
Different formats	Jordan / jordan / JO; different time zones	One spelling, one time zone (UTC)
Outlier	500 failed logins in one minute	Check first: error or attack?

In security, the outlier is often the attack: never delete it blindly

Practice Datasets

Phishing Email Detection (Kaggle)
18,650 e-mails labelled phishing or safe (Case 1)
https://www.kaggle.com/datasets/subhajournal/phishingemails

Network Intrusion Detection (Kaggle)
25,192 network connections labelled normal or anomaly (Case 4)
https://www.kaggle.com/datasets/sampadab17/network-intrusion-detection

Credit Card Fraud Detection (Kaggle)
284,807 payments, only 492 frauds (Case 7)
https://www.kaggle.com/datasets/mlg-ulb/creditcardfraud

Loghub: OpenSSH logs (GitHub)
2,000 real SSH log lines with many failed logins (logs, Case 5)
https://github.com/logpai/loghub/tree/master/OpenSSH

Class Activity: Match the Case

For each situation: which case is it, and which kind of learning helps?

1

E-mail from “ajloun-nationa1.com” asks for your password

2

A PC suddenly encrypts 400 files a minute

3

One account logs in from Amman and Tokyo 5 minutes apart

4

One address contacts 1,000 ports in 3 seconds

5

An employee copies 5,000 files before resigning

6

A card pays 1,900 JD abroad at 03:40

7

Which of 300 new CVEs should we patch first?

8

A video call with a “manager” asking for a transfer

Key Terms

Term	Definition
Cybersecurity	Protecting systems, networks and data from attack and unauthorised access
CIA triad	Confidentiality, integrity, availability
Threat / vulnerability	Something that can cause harm / a weakness
Attack / risk	An attempt to use a weakness / likelihood × impact
Event / alert / incident	Recorded happening / flagged event / confirmed problem
SOC / SIEM	Monitoring team / system that collects logs and alerts
Phishing / malware	Fake trusted message / harmful software
Ransomware / intrusion	Encrypts files for payment / unauthorised access
Account takeover / insider	Attacker controls a real account / misuse from inside
Deepfake	Fake audio, video or image made with AI
Log / data processing	Record of an event / steps from raw data to a clean table

Review Questions

- Define the CIA triad and give an attack that breaks each part.
- What is the difference between a threat, a vulnerability and an attack?
- Explain events, alerts and incidents. Why does a SOC need AI to move from one to the next?
- How does AI detect malware that signature antivirus misses?
- Why is impossible travel a sign of account takeover? What else could explain it?
- Name the five data-processing steps from logs to data.

Summary

- Cybersecurity protects confidentiality, integrity and availability
- Events → alerts → incidents; the SOC uses a SIEM
- AI fights phishing, malware, ransomware, intrusions, account takeover, insiders and fraud
- Generative AI helps analysts, and attackers use it too
- Every case starts with data, mostly logs: collect, parse, clean, turn into numbers, split

Challenge Question (Optional)

For students who want to go further

The university SOC can buy only one AI tool this year: an AI phishing filter or an AI login-behaviour monitor (UEBA).

Which would you buy first, and why?

Which logs or data would the tool need from the university?